

Problem Description

Given an array A of N integers. Now, you have to output the sum of unique values of the maximum subarray sum of all the possible subarrays of the given array A .

Note: Subarray means contiguous elements with at-least one element in it.

Constraints

$$1 \leq N \leq 2000$$

$$0 \leq |A_i| \leq 10^9$$

Input Format

The first line of the input contains a single integer N , the total number of elements in array A .

The next line of the input contains N space-separated integers representing the elements of the array.

Output Format

The only single line of the output should contain a single integral value representing the answer to the problem.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

4
5 -2 7 -3

EXPECTED OUTPUT

17

Test Case 2

INPUT (STDIN)

15
2 3 4 5 6 7 3 4 2 7 6 8 6 4 9

EXPECTED OUTPUT

2021

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
int NA[N];
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

555

Test Case 3

NLOC

73